American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

CN Lecture 01: Networking Basics (TCP/IP Model)

OBE Based Questions and Answers

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1. **A company wants to develop a web-based file sharing system. Explain how data travels from the sender to the receiver using the TCP/IP protocol suite.**

   Answer:

   The TCP/IP protocol suite consists of five layers that work together to ensure successful communication.

   (a) **Application Layer**
   The application layer receives data from the user's application. Protocols such as HTTP, FTP, DNS, and SMTP operate at this layer.

   (b) **Transport Layer**
   The transport layer divides the message into segments and provides reliable (TCP) or unreliable (UDP) communication. It also performs port addressing, error control, flow control, and congestion control.

   (c) **Network Layer**
   The network layer adds IP addresses, performs routing, and converts segments into IP packets.

   (d) **Data Link Layer**
   The data-link layer adds MAC addresses, converts packets into frames, and performs error detection.

   (e) **Physical Layer**
   The physical layer converts frames into electrical or optical signals and transmits them through the communication medium.

   At the receiver, the process is reversed through decapsulation until the original message reaches the destination application.

2. **Compare TCP and UDP. Recommend the most suitable protocol for video streaming and online banking with proper justification.**

   Answer:

Feature	**TCP**	**UDP**
Connection	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Speed	Slower	Faster
Error Recovery	Available	Not Available
Applications	Banking, Email, FTP	Video Streaming, Gaming, VoIP

   TCP should be used for online banking because reliable data delivery is essential.

   UDP should be used for video streaming because speed is more important than occasional packet loss.

3. **A network administrator needs to connect several computers inside a single office. Compare Hub, Bridge, Switch, and Router and recommend the most appropriate device.**

   Answer:

Device	**Layer**	**Address Used**	**Main Function**
Hub	Layer 1	None	Broadcasts signals to every port
Bridge	Layer 2	MAC Address	Filters frames between segments
Switch	Layer 2	MAC Address	Sends frames only to intended device
Router	Layer 3	IP Address	Connects multiple networks

For connecting computers inside one office, a Switch is the best choice because it provides dedicated bandwidth, reduces collisions, and forwards frames only to the intended destination.

4. **Analyze why a switch performs better than a hub in modern computer networks.**

   Answer:

   A switch performs better than a hub because:

   - A switch forwards frames only to the destination device.
   - A hub broadcasts data to every connected device.
   - Each switch port has its own collision domain.
   - A hub has one shared collision domain.
   - A switch provides dedicated bandwidth.
   - A switch supports hardware-based forwarding, making it much faster.

   Therefore, switches improve network performance, efficiency, and security.

5. **Explain the concept of Collision Domain and Broadcast Domain with suitable examples.**

   Answer:

   A collision domain is a part of the network where packet collisions can occur when multiple devices transmit simultaneously.

   A broadcast domain is the portion of a network where broadcast packets can reach every device.

Device	**Collision Domain**	**Broadcast Domain**
Hub	One shared domain	One domain
Switch	One per port	One domain
Bridge	One per port	One domain
Router	One per interface	One per interface

   Routers separate broadcast domains, whereas switches mainly reduce collision domains.

6. **A company wants to upgrade its network from Fast Ethernet to Gigabit Ethernet. Compare the Ethernet standards and recommend a suitable technology.**

   Answer:

Standard	**Speed**	**Cable**	**Maximum Length**
10BASE-T	10 Mbps	Cat3	100 m
100BASE-T	100 Mbps	Cat5	100 m
1000BASE-T	1000 Mbps	Cat5e	100 m
10GBASE-T	10 Gbps	Cat6/Cat6a	100 m

   For most organizations, 1000BASE-T is recommended because it provides high speed while using affordable Cat5e or Cat6 cables.

7. **Evaluate the evolution of WLAN standards from IEEE 802.11 to IEEE 802.11ac.**

   Answer:

   Wireless LAN standards have continuously improved in speed, bandwidth, modulation techniques, and MIMO support.

Standard	**Frequency**	**Maximum Speed**	**Remark**
802.11	2.4 GHz	2 Mbps	First WLAN standard
802.11b	2.4 GHz	11 Mbps	Improved speed
802.11a	5 GHz	54 Mbps	Less interference
802.11g	2.4 GHz	54 Mbps	Backward compatible
802.11n	2.4/5 GHz	600 Mbps	Introduced MIMO
802.11ac	5 GHz	6.93 Gbps	High-speed wireless networking

The evolution shows continuous improvements in data rate, wireless efficiency, and communication reliability.

8. **A network consists of routers, switches, and hubs. Analyze how collision domains and broadcast domains change when each device is added.**

   Answer:

   - Adding a Hub increases the number of connected devices but all devices remain in one collision domain and one broadcast domain.
   - Adding a Switch creates one collision domain for each port while keeping a single broadcast domain.
   - Adding a Router separates both collision domains and broadcast domains because each router interface forms an independent network.

   Therefore, routers provide the highest level of network segmentation, while switches mainly reduce collisions.

A software company is developing an online examination system. Explain how each layer of the TCP/IP protocol suite contributes to transmitting a student's answer from the client computer to the server.

Answer:

When a student submits an answer, the TCP/IP protocol suite performs the following operations:

1. **Application Layer**
   The examination software generates the message and uses protocols such as HTTP or HTTPS to communicate with the server.
2. **Transport Layer**
   The message is divided into segments. TCP ensures reliable delivery through error detection, acknowledgments, flow control, and congestion control.
3. **Network Layer**
   Each segment is encapsulated into an IP packet. The source and destination IP addresses are added, and routers determine the best path.
4. **Data Link Layer**
   Each packet is converted into a frame by adding source and destination MAC addresses. Error detection is performed using frame checking.
5. **Physical Layer**
   Frames are converted into electrical or optical signals and transmitted through cables or wireless media.

At the destination, the reverse process (decapsulation) reconstructs the original message.

A company wants reliable communication for financial transactions but fast communication for live video conferencing. Recommend suitable transport layer protocols for both cases with justification.

Answer:

Financial transactions require **TCP** because data accuracy is more important than speed. TCP guarantees reliable delivery through acknowledgments, retransmissions, sequencing, and flow control.

Live video conferencing should use **UDP** because low latency is more important than perfect reliability. UDP avoids retransmissions and acknowledgments, resulting in faster communication.

Therefore,

- Financial Transactions → TCP
- Video Conferencing → UDP

Analyze the advantages and disadvantages of using a Hub instead of a Switch in a Local Area Network (LAN).

Answer:

Advantages of Hub

- Low cost.
- Easy installation.
- Suitable for very small networks.

Disadvantages

- Broadcasts data to every device.
- Creates one collision domain.
- Shares bandwidth among all devices.
- Lower performance.
- Poor security.

A switch is preferred because it forwards data only to the destination device and provides dedicated bandwidth.

Compare Bridge and Switch. Which device is more suitable for a modern enterprise network? Justify your answer.

Answer:

Both Bridge and Switch operate at Layer 2 and use MAC addresses for forwarding.

Feature	Bridge	Switch
OSI Layer	Layer 2	Layer 2
Ports	Usually 2–4	Up to hundreds
Forwarding	Software	Hardware ASIC
Speed	Slower	Faster
Bandwidth	Limited	Dedicated per port

A switch is more suitable because it supports more ports, provides higher speed, and offers better bandwidth utilization.

An organization has multiple branch offices located in different cities. Which networking device should be used to connect these networks? Explain your answer.

Answer:

A **Router** should be used.

Reasons:

- Operates at Layer 3.
- Uses IP addresses for forwarding.
- Connects multiple independent networks.
- Determines the best routing path.
- Supports different network technologies.

Therefore, routers are essential for Wide Area Networks (WANs).

Explain why routers separate broadcast domains while switches do not.

Answer:

Switches forward broadcast frames to every port within the same LAN. Therefore, all switch ports remain in one broadcast domain.

Routers do not forward Layer-2 broadcast packets between interfaces. Each router interface creates a separate broadcast domain.

Consequently,

- Switch: One broadcast domain.
- Router: Separate broadcast domain per interface.

Analyze the encapsulation and decapsulation process in the TCP/IP protocol suite.

Answer:

During encapsulation,

1. Application creates the message.
2. Transport adds TCP/UDP header forming a Segment.
3. Network adds IP header forming a Packet.
4. Data Link adds MAC header and trailer forming a Frame.
5. Physical layer converts the frame into signals.

During decapsulation, headers are removed in reverse order until the application receives the original message. Encapsulation enables communication between heterogeneous devices using standard protocols.

A network administrator wants to minimize collisions in a busy office network. Recommend a suitable networking device and justify your recommendation.

Answer:

The recommended device is a **Switch**.

Reasons:

- Each port forms an independent collision domain.
- Supports full-duplex communication.
- Reduces packet collisions.
- Provides dedicated bandwidth.
- Improves overall network efficiency.

Therefore, switches significantly improve LAN performance.

Compare Ethernet standards from 10BASE-T to 10GBASE-T and explain the technological improvements.

Answer:

Standard	Speed	Cable	Maximum Distance
10BASE-T	10 Mbps	Cat3	100 m
100BASE-T	100 Mbps	Cat5	100 m
1000BASE-T	1 Gbps	Cat5e	100 m
10GBASE-T	10 Gbps	Cat6/Cat6a	100 m

The major improvements include higher bandwidth, improved cable quality, increased transmission speed, and better support for modern network applications.

Evaluate the advantages of IEEE 802.11ac compared to IEEE 802.11b.

Answer:

IEEE 802.11ac offers several advantages:

- Operates on the less congested 5 GHz band.
- Supports MIMO technology.

- Uses 256-QAM modulation.
- Provides bandwidth up to 160 MHz.
- Achieves data rates up to 6.93 Gbps.

Compared to IEEE 802.11b (11 Mbps), IEEE 802.11ac provides significantly higher speed, capacity, and wireless efficiency.

A university wants to build a high-speed wireless campus network. Recommend an appropriate WLAN standard and justify your recommendation.

Answer:

IEEE 802.11ac is recommended because:

- Supports very high data rates.
- Operates in the 5 GHz frequency band.
- Supports MIMO technology.
- Uses wider channels (80/160 MHz).
- Suitable for high-density wireless users.

Therefore, IEEE 802.11ac provides excellent performance for modern campus environments.

Evaluate the importance of connecting devices (Repeater, Hub, Bridge, Switch, and Router) in building an efficient computer network.

Answer:

Each connecting device performs a unique function.

- Repeater extends transmission distance.
- Hub connects devices but broadcasts to all ports.
- Bridge filters traffic using MAC addresses.
- Switch provides efficient Layer-2 communication with dedicated bandwidth.
- Router connects multiple networks using IP addresses.

Together, these devices improve communication efficiency, reduce congestion, increase scalability, and enable communication across local and wide area networks.